

# Resolution Calibration Standar

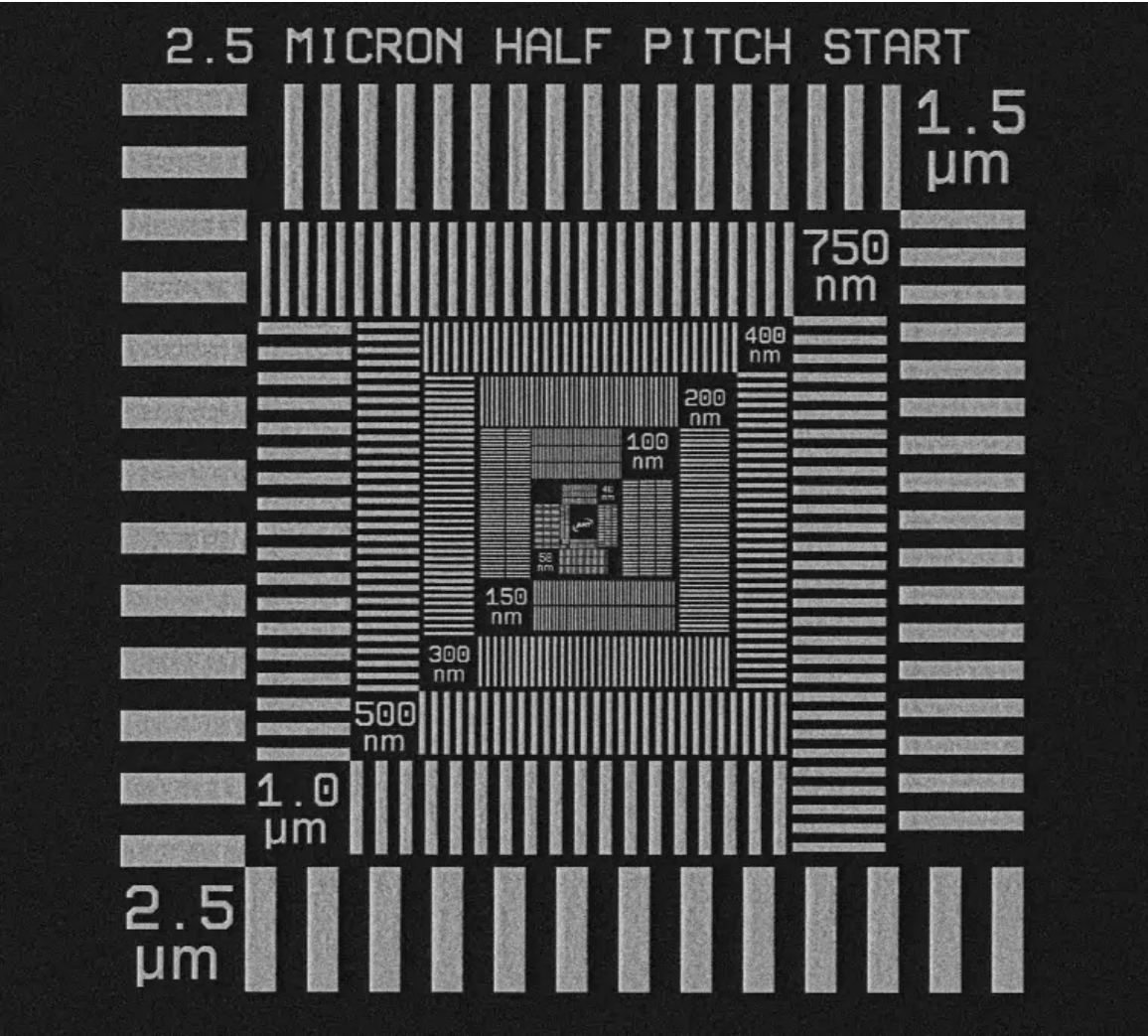
EUV, X-Ray and visible light calibration standards

## Customizable High-Resolution Calibration Standards

The ANT calibration standard was designed for the highest resolution requirements of X-Ray imaging systems with nanoscale features. Available for both soft and hard X-Ray regimes, EUV and optical setups.

|                    |
|--------------------|
| <b>Nested L's</b>  |
| <b>Siemen Star</b> |
| <b>Nano USAF</b>   |
| <b>Gratings</b>    |

These varying line and spaces are useful for inspection and determining the resolution of a system.



|                 |
|-----------------|
| Hard X-rays     |
| Soft X-rays     |
| TEM             |
| Custom Work     |
| Custom Frames   |
| Standard Holder |

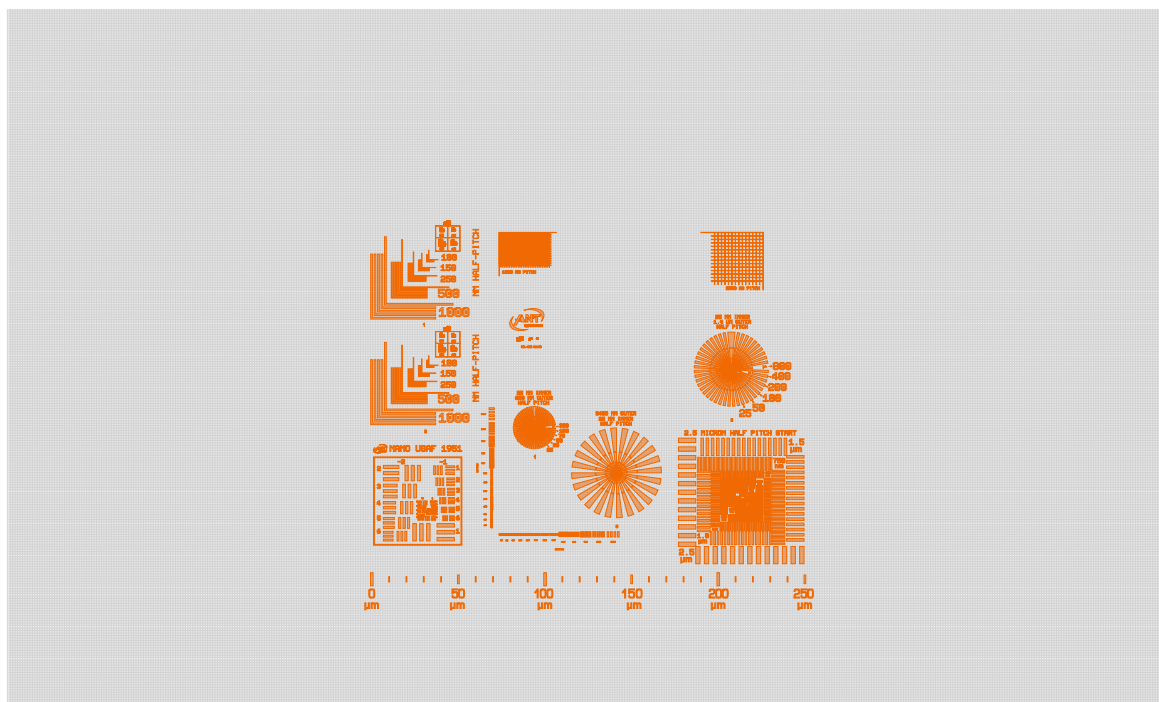
25 nm Design Layout (Other designs available with 12.5, 15, 100, and 200 nm, etc

min. feature sizes)

Toggle Scale Unit (Currently: nm)

Enable Measure Mode

Re-center



200000 nm

Cursor: X = 125574.6 nm, Y = -65467.7 nm

## Options (for different energies)

| Device                                                                                                                                                           | Target Energies             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>EUV Free-standing silicon nanoUSAF1951 (&gt;100 nm Si<sub>3</sub>N<sub>4</sub>)<br/>(negative polarity, etched into Si<sub>3</sub>N<sub>4</sub> membrane)</b> | <b>10-90 eV (124-14 nm)</b> |
| <b>Ultra-High Resolution Soft X-Ray Calibration Standard (70 nm Au)<br/>(15 nm half pitch)</b>                                                                   | <b>100 - 500 eV</b>         |
| <b>Soft X-Ray Calibration Standard (150 nm Au)<br/>(25 nm half pitch or better)</b>                                                                              | <b>100 - 1000 eV</b>        |
| <b>Hard X-Ray Calibration Standard (&gt;500 nm Au)<br/>(40 nm half-pitch or better)</b>                                                                          | <b>100 - 12000 eV</b>       |

### Included Features:

- Nested L's
- Siemen star patterns (large and small)
- Varying pitch gratings and mesh
- Nano USAF 1951

| Custom Options                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Custom Logo (Max 50 <math>\mu\text{m}^2</math> area)</b>                                                                             |
| <b>Low stress 50 nm silicon nitride membrane</b>                                                                                        |
| <b>Low stress 200 nm silicon carbide membrane</b>                                                                                       |
| <b>Fused Silica Substrate (~500 <math>\mu\text{m}</math> thick)<br/>•Chrome metal on glass (positive tone-only) &gt;60 nm thickness</b> |

**Negative polarity (features transparent)**

- 40 nm minimum feature size (soft X-ray)
- 80 nm minimum feature size (hard X-ray)

**Custom chip sizes**

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## X-Ray Optics

Zone Plates (<https://www.appliednt.com/zone-plates/>)

Resolution Calibration Standards (<https://www.appliednt.com/calibration-standards/>)

Gratings (<https://www.appliednt.com/gratings/>)

Holders (<https://www.appliednt.com/holders/>)

Zone Plate Calculator (<https://www.appliednt.com/zpcalc/>)

Online GDS Viewer (<https://www.appliednt.com/gds-viewer/>)

## Integrated Photonics

Introduction to NanoSOI Fabrication Service (<https://www.appliednt.com/nanosoi-fabrication-service/>)

NanoSOI Design Center (<https://www.appliednt.com/nanosoi/sys/>)

Measurement, Design, and Packaging (<https://www.appliednt.com/other-services/>)

## Additional links

Careers (<https://www.appliednt.com/careers/>)

Contact (<https://www.appliednt.com/contact/>)



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